LESSON 4.5

Determine the Whole Given the Part and Percent

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can determine the whole when I know a part and the percent that part represents.

WHY THIS MATTERS Every percent problem has three pieces: the part, the whole, and the percent. So far the whole has always been handed to you. Today it is the missing piece — a sale price without the original price, a car's current value without its sticker price. Your job is to work backwards to the whole. For example, Akela paid $42 for a sweater on sale, and that was 60% of the original price. What was the original price?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Part *Spanish: Parte*	The amount you already know — the piece of the whole that the percent describes.	$42 is the sale price
Whole *Spanish: Total*	The full amount the percent is measured against. The whole is always 100%.	the original price
Double number line *Spanish: Recta numérica doble*	Two lines lined up — dollars on one, percents on the other — so matching marks show the same amount two ways.	$0, $7, $70 · 0%, 10%, 100%
Equation *Spanish: Ecuación*	A math sentence with an equal sign. Here it says percent × whole = part, with the whole unknown.	0.7v = 26,600 — the percent times the unknown whole equals the part
Percent *Spanish: Porcentaje*	A ratio comparing a number to 100. To compute with it, write it in decimal form: 60% = 0.6.	60% = 0.6 · 70% = 0.7
Inverse operation *Spanish: Operación inversa*	The operation that undoes another operation.	Division undoes multiplication.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

24 stickers are 40% of a pack. How many stickers are in the pack?

1. 40% ÷ 4 = 10%, so 24 ÷ 4 = 6.
2. 10% × 10 = 100%, so 6 × 10 = 60.
3. Check: 0.40 × 60 = 24.

Answer: 60 stickers

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Sheng's car is now worth $26,600, which is 70% of what his parents paid for it. Find the original price.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The part is _____ and it matches _____ %.
- Ten percent of the whole is _____ .
- The whole is _____ because _____ .
- I checked my answer by finding _____ % of it.

Word bank part | whole | percent | double number line | equivalent ratio | scale | divide | multiply | ten percent | one hundred percent | table | check

WATCH OUT!

A common mistake

A common mistake when finding the whole is multiplying by the percent instead of dividing. For "$42 is 60% of the original price," computing 0.6 × 42 = $25.20 makes the original price SMALLER than the sale price

LESSON 4.5

Determine the Whole Given the Part and Percent

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each number by the role it plays in "$42 is 60% of the original price."

Groups: The part | The percent | The unknown whole

_____ $42

_____ 60%

_____ the original price

_____ the amount Akela paid

_____ 0.6

_____ 100%

HINT Which number was actually paid?

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

$20 is 50% of what number?

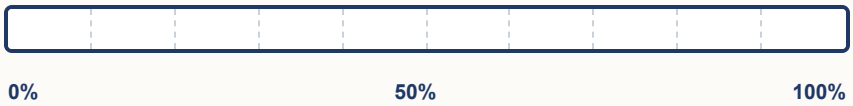
(A) $10

(B) $25

(C) $40

(D) $70

HINT 50% means half of the whole.



WORK

1. **Whole** *100% is what number?*
2. **Find 10%** *Divide by 10.*
3. **Build** *Add or multiply to the percent.*
4. **Check** *Is it under or over the whole?*

3 **Circle the letter of the best answer. Show how you know.**

Which equation says "30 is 25% of some number n"?

(A) 0.25n = 30

(B) n = 0.25 × 30

(C) 25n = 30

(D) n = 30 + 25

HINT Which number is the part, and which is the percent?

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

$42 is 60% of the original price. Should the original price be more or less than $42?

(A) More, because 60% is less than the whole

(B) Less, because 60 is a big number

(C) Exactly $42

(D) You cannot tell

HINT What percent stands for the whole?

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

75 minutes is 15% of the school day. How long is the school day?

(A) 11.25 minutes

(B) 90 minutes

(C) 500 minutes

(D) 1,125 minutes

HINT 15% is three groups of 5%. Split 75 minutes into 3 equal groups to find 5%.



WORK

1. **Whole** *100% is what number?*
2. **Find 10%** *Divide by 10.*
3. **Build** *Add or multiply to the percent.*
4. **Check** *Is it under or over the whole?*

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Find the error

1. **Setup** $42 is 60% of the original price. Find the original price.
2. **Step 1** 60% = 0.6
3. **Step 2** 0.6 × 42 = 25.20
4. **Step 3** The original price was $25.20.

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT Should the original price be more or less than $42?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How can you tell, before doing any arithmetic, whether the answer should be bigger or smaller than the number you were given?

SENTENCE STARTER If the percent is less than 100%, the given number is a ___, so the whole must be ___ than it.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 4.5

Determine the Whole Given the Part and Percent

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
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PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

Which statement correctly names the pieces of "18 students is 30% of the class"?

(A) 18 is the whole, 30% is the part

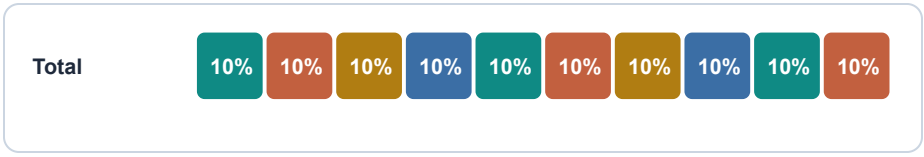
(B) 30 is the part, 18 is the percent

(C) The class size is the part

(D) 18 is the part, 30% is the percent, the class size is the whole

SHOW YOUR WORK

2 **Use the model to solve. Show your work.**



A sweater's sale price of $42 is 60% of the original price. Each section is 10%. Use the bar model to find the original price.

SHOW YOUR WORK

Answer: _____

3 **Complete the table. Show how you found each value.**

$42 is 60% of the original price. Complete the ratio table to reach 100%.

Percent	Dollars
60%	42
10%	
100%	

SHOW YOUR WORK

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

18 students are 30% of an after-school group. How many students are in the whole group?

(A) 6
(B) 30
(C) 60
(D) 180



0% 50% 100%

WORK

5 **Circle the letter of the best answer. Show how you know.**

A car is now worth $26,600, which is 70% of its purchase price. What was the purchase price?

(A) $18,620
(B) $26,670
(C) $38,000
(D) $45,200



0% 50% 100%

WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Find the error

1. **Setup** 18 students is 30% of the class. How many students are in the class?
2. **Step 1** 30% = 0.3
3. **Step 2** 18 ÷ 0.3 = 60
4. **Step 3** There are 60% of students in the class.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How can you tell, before doing any arithmetic, whether the answer should be bigger or smaller than the number you were given?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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Challenge · Extension

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Each row gives a part and a percent. Find the whole.

Part	Percent	Whole
$42	60%	70
$26,600	70%	
75 min	15%	
18 students	30%	

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

2 **Find the mistake. Explain it, then show the correct work.**

Find the error

1. **Setup** A jacket is on sale for $60, which is 80% of the original price. Find the original price.
2. **Step 1** The discount is 20%.
3. **Step 2** 20% of 60 = 12.
4. **Step 3** The original price was 60 + 12 = $72.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 **Answer in complete sentences.**

A toy store's puzzle stock was 25% 300-piece, 15% 500-piece, and the rest 1,000-piece. During a sale the store sold ALL of its 300-piece and 500-piece puzzles — 120 puzzles in total. How many of each type did the store have before the sale? Explain your reasoning.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

How can you tell, before doing any arithmetic, whether the answer should be bigger or smaller than the number you were given?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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